Phase 1 Wall-Time Scaling with Number of Regions

Empirical Measurements, Power-Law Fit, and the Path to $N > 100$

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Abstract

This document analyses how Phase 1 (PGD relaxation) wall time grows with the number of regions N. Two pre-optimization data points — $N = 10$ (1.2 h, 7 levels, torus, random init) and $N = 30$ (20.7 h–38.2 h, 5 levels, torus, seeded init) — yield an empirical power-law exponent of approximately $\alpha \approx 3.1$, meaning wall time scales roughly as N^3. Since these measurements were taken, the serial optimizations of §6 (the backtracking step warm-start, the projection inner-loop cleanup, and post-step gradient reuse) landed; they are *constant-factor* wins ($\approx$3.8× at $N = 50$) and so do **not** change α, but they re-anchor every absolute estimate below. A profiled post-optimization run measures $N = 50$ at **11.6 h** (clean host, best seed, 5 levels). Anchored on that measured point with $\alpha = 3.1$, the projections become $\sim$**4 days** for $N = 100$ and $\sim$**14 years** for $N = 1000$ — the latter still infeasible at scale on a single machine, though an order of magnitude below the pre-optimization $\sim$120 yr. The dominant cost remains the orthogonal projection, whose share fell from 88–90% to $\approx$80% after the inner-loop cleanup and which still scales as $\mathcal{O}(N)$ per call. The N^3 overall scaling arises because both the per-call cost and the number of iterations grow with N. The remaining levers — the durable algorithmic fix is a closed-form simplex projection with a dual-Newton equal-area coupling (§7), plus parallelising across regions — are required before the method can scale to $N > 100$. Inner-solver dual warm-start, listed as future work in an earlier draft, has since been ruled out on correctness grounds.

Contents

1 Context and Motivation

The long-term goal of the surface-partition project is to compute minimal-perimeter partitions for $N \gg 30$ regions, with $N = 1000$ as a target. Document 04 (*Phase 1 Timing Profile*) established that 88–90% of Phase 1 wall time is spent in the orthogonal projection $\Pi_{\mathcal{C}}$ onto the partition-of-unity and equal-area constraints, and that the projection cost per call is $\mathcal{O}(\mathrm{mPII} \cdot VN)$, where V is the number of mesh vertices and mPII is the mean inner-iteration count.

This document quantifies how total Phase 1 wall time actually grows with N using the N values for which profiled data exist, projects the fit to larger N, and identifies which factors in the cost model are responsible for the steep scaling.

Remark 1.1 (Optimization status)**.** The $N = 10$ and $N = 30$ runs of §3 predate the Phase 1 PGD serial optimizations (Changes A/B/C; see §6 and `docs/reference/PHASE1_PGD_SERIAL_OPTIMIZATION_AUDIT.m`
The $N = 50$ run of §3.3 is post-optimization. Because Changes A/B/C are constant-factor wins, the *exponent* α derived from the pre-optimization pair is unchanged, but the *absolute* projections of §5 are re-anchored on the measured post-optimization $N = 50$ point.

2 Notation and Setup

Throughout this document:

- N — number of regions (partition cells).
- V — number of mesh vertices at the finest refinement level.
- L — number of refinement levels.
- T — total Phase 1 wall time (seconds or hours).
- mBT — mean Armijo backtrack trial steps per major iteration.
- mPII — mean inner-iteration count per projection call.
- K — total major (outer) PGD iterations summed across all levels.

The cost model from document 04 is

$$T \;\sim\; \underbrace{K \cdot (\mathrm{mBT} + 1)}_{\#\text{ projection calls}} \;\times\; \underbrace{\mathrm{mPII} \cdot VN}_{\text{cost per call}}. \tag{1}$$

3 Measured Data Points

3.1 $N = 10$ reference run

An early profiled run used the random initialisation, 7 refinement levels, and an 80×66 initial mesh growing to $V_{10} = 187{,}128$ vertices at the finest level. The total wall time was $T_{10} = 1.19\,\mathrm{h}$. Per-level iteration counts were not recorded in this run's metadata (older schema), so the iteration breakdown is unavailable; only the aggregate time is used.

Table 1: $N = 10$ reference run (torus, $\lambda = 3.25$, seed 13001502, random init).

Parameter	**Value**
Regions N	10
Refinement levels L	7
Initial mesh	80×66 (5 280 vertices)
Final mesh V_{10}	187 128 vertices
Total wall time T_{10}	1.19 h
Initialisation	random

3.2 $N = 30$ sweep (seeded initialisation)

Six profiled runs used the seeded (Voronoi farthest-point) initialisation, 5 refinement levels, and a 90×86 initial mesh growing to $V_{30} = 107{,}484$ vertices. All six produced valid 30-region partitions (no dormant cells). Wall times vary by a factor of 1.8× across seeds due to line-search behaviour at intermediate levels; the best-case run (seed 84172851) is used as the lower bound for scaling estimates, and the worst-case run (seed 61847392) as the upper bound.

Table 2: $N = 30$ profiled runs (torus, $\lambda = 2.1$, 5 levels, seeded init, $V_{30} = 107{,}484$ vertices).

Seed	Wall (h)	Iters K	mBT	mPII	proj %
84172851	20.74	27 001	—	19.4	89.4
83920164	25.86	38 574	—	19.5	90.1
47301856	29.09	37 980	—	17.4	89.2
13151129	30.14	45 110	—	16.4	88.1
61847392	38.17	49 811	—	17.6	89.0
min / max	**20.74 / 38.17**				

Table 3 shows the per-level projection cost for the best-case $N = 30$ run, demonstrating the $\mathcal{O}(VN)$ trend within a single run as V grows across levels.

Table 3: Per-level projection call timing for the $N = 30$ reference run (seed 84172851). Mean projection time grows with V as expected.

Level	Vertices V	Proj calls	Mean proj (ms)	Level wall (s)
0	7 740	1 600	38.9	77
1	21 888	57 935	182.0	2 995
2	43 228	42 880	196.9	4 847
3	71 760	34 080	571.2	8 823
4	107 484	32 627	865.3	27 934

3.3 $N = 50$ measured run (post-optimization)

A profiled run taken *after* the serial optimizations of §6 provides the first real $N = 50$ data point and the new anchor for absolute projections. It used the seeded initialisation, 5 refinement levels, and a 100×96 initial mesh growing to $V_{50} = 114{,}144$ vertices. On a clean host (best seed 84172851) the total wall time was $T_{50} = 41{,}754\,\mathrm{s} = \mathbf{11.6}\,\mathrm{h}$.

Table 4: $N = 50$ post-optimization run (torus, $\lambda = 2.1$, seed 84172851, 5 levels, seeded init, $V_{50} = 114{,}144$ vertices; run `run_20260627_234511`).

Quantity	**Value**
Total wall time T_{50}	11.6 h (41 754 s)
Major iterations K	54 154
Mean backtracks mBT	1.93
Mean inner iters mPII	13.76
Projection share	80.5 % of wall
Energy / gradient share	8.2 % / 7.4 %
Backtrack share (net)	2.5 %

Compared against the pre-optimization regime quoted for $N = 30$ (Table 2; mPII $\approx$ 16–20, projection share 88–90%, mBT in the 4–12 range at the tail), the optimization effects are directly visible in the profile:

- mBT collapsed to 1.93 — Change A's line-search warm-start removed most of the rejected backtrack trials (and with them the rejected-step projections), which is the largest single contributor to the $\approx 3.8\times$ speedup.
- the projection share fell from 88–90% to 80.5% — Change B trimmed the per-inner-iteration $\mathcal{O}(VN)$ waste, and the rejected-step projections eliminated by Change A no longer inflate the bucket.
- mPII is 13.8, modestly below the pre-optimization 15–20; Change B did not target the inner-iteration *count* (it left the algorithm result-preserving), so this is mostly run-to-run variation, not a warm-start effect.

Remark 3.1 (Hardware variance dominates absolute timing)**.** The same config and seed took 43.9 h on a contended host (`run_20260625_113015`) versus 11.6 h on a clean one — a $3.8\times$ spread from hardware contention alone, comparable to the optimization speedup itself. Absolute projections below are anchored on the fastest clean run; worst-seed $\times$ contended-host combinations can run 2–4$\times$ higher.

4 Empirical Scaling Fit

4.1 Mesh-size normalisation

The two data points use different final mesh sizes ($V_{10} = 187{,}128$ and $V_{30} = 107{,}484$). Since projection cost scales linearly with V, we normalise T_{10} to the $N = 30$ mesh before fitting:

$$T_{10}^{*} = T_{10} \times \frac{V_{30}}{V_{10}} = 1.19 \times \frac{107{,}484}{187{,}128} \approx 0.683\,\mathrm{h}. \tag{2}$$

4.2 Power-law exponent

Assuming $T \propto N^{\alpha}$ and using the adjusted pair $(N, T) \in \{(10, 0.683), (30, 20.74)\}$ (hours):

$$\alpha = \frac{\ln(T_{30}/T_{10}^{*})}{\ln(30/10)} = \frac{\ln(30.4)}{\ln(3)} \approx \frac{3.41}{1.10} \approx 3.1. \tag{3}$$

The exponent $\alpha \approx 3$ decomposes into two factors:

- $\mathcal{O}(N)$ **per projection call** — the inner loop of $\Pi_{\mathcal{C}}$ operates on the $V \times N$ density matrix, so each call costs linearly in N.
- $\mathcal{O}(N^2)$ **iteration count** — more regions produce a more crowded energy landscape; empirically, K grows roughly as N^2 between the two data points (from $\sim$3,500 at $N = 10$, estimated, to $\sim$27,000–50,000 at $N = 30$, measured).

Combined: $T \propto N \cdot N^2 = N^3$, consistent with $\alpha \approx 3.1$.

Remark 4.1 (Uncertainty in α)**.** The fit rests on only two data points from runs with different initialisations (random vs. seeded) and different level counts (7 vs. 5). The true exponent could plausibly lie in the range [2.5, 3.5]. The estimates below should be treated as order-of-magnitude guides, not precise predictions.

Remark 4.2 (Why the optimizations do not move α)**.** The serial optimizations (§6) multiply T by a roughly N-independent constant ($\approx$1/3.8), which cancels in the ratio T_{30}/T^*_{10} of Eq. (3). The exponent is therefore still $\alpha \approx 3.1$. What *did* change is the absolute scale: §5 anchors $T(N) = T_{50}\,(N/50)^{\alpha}$ on the measured post-optimization point $T_{50} = 11.6\,\mathrm{h}$, rather than extrapolating from the pre-optimization $N = 30$ table. As a consistency check, the pre-optimization fit predicts $T_{50} \approx 20.74\,(50/30)^{3.1} \approx 102\,\mathrm{h}$; dividing by the measured 3.8× speedup gives $\approx$27 h, still $\sim$2× above the measured 11.6 h — within the order-of-magnitude tolerance the two-point fit warrants.

5 Projections to Larger N

Anchoring on the measured post-optimization point with $T(N) = T_{50} \cdot (N/50)^{\alpha}$, $T_{50} = 11.6\,\mathrm{h}$ (clean host, best seed) and $\alpha = 3.1$:

Table 5: Projected Phase 1 wall-time estimates (single clean machine, same 5-level torus mesh schedule), anchored on the measured post-optimization $N = 50$ point. The "contended" column applies the empirical 3.8× hardware/worst-seed penalty of §3.3 as a practical upper bound. The $N = 50$ row is measured, not projected.

N	$(N/50)^{3.1}$	Clean (h)	Clean (days)	Contended (days, ×3.8)
50	1.0	11.6 *(meas.)*	0.5	1.8
75	3.5	41	1.7	6.4
100	8.6	99	4.1	15.7
200	73.5	853	35.5	135
1 000	$\approx 1.1 \times 10^4$	$\approx 1.25 \times 10^5$	≈5,200	$\approx 2.0 \times 10^4$

At $N = 1000$ the clean-host estimate is $\sim 1.25 \times 10^5\,\mathrm{h} \approx 14$ years; the contended bound is $\sim$54 years. The practical conclusions:

- $N = 50$ is **routinely feasible** on a single machine — measured at half a day on a clean host with the seeded initialisation (an order of magnitude below the pre-optimization projection of Table 5's earlier draft).
- $N = 100$ is **feasible on a single clean machine** ($\sim$4 days), and comfortable on a cluster with multiple seeds in parallel.
- $N = 200$ is a **multi-week** single-machine run; practical only with the algorithmic fix below or substantial parallelism.

- $N = 1000$ remains **not achievable** with the current algorithm — ~ 14 years even on a clean host. The serial optimizations bought roughly one order of magnitude (from ~ 120 yr) but not the three-to-four orders needed.

6 Implemented: Serial Optimizations (Changes A/B/C)

The optimizations below have *landed* (merge `bae8cd2`); they are the source of the $\approx 3.8\times$ constant-factor speedup measured at $N = 50$ (§3.3). They preserve the optimizer's output — the $N = 50$ partition was validated bit-identical to the pre-optimization result (contour perimeter 152.3883843639, 100% winner-take-all agreement) — so they reduce the constant in Eq. (1) without changing α. The full audit (#1–#13, with measured impact tiers) is recorded in `docs/reference/PHASE1_PGD_SERIAL_OPTIMIZATION_AUDIT.md`.

Change A — backtracking step warm-start (audit #1, #8). `step0` was hard-reset to 1.0 at the top of every PGD iteration, so each iteration re-walked the Armijo backtrack from scratch; near convergence the accepted step decayed to $\sim 2^{-8}$, i.e. ~ 8 rejected trials per iteration, each a full projection + full energy. Change A carries the last accepted step and starts the next search from $\min(s_{\max}, s_{\text{last}}/\rho)$ (Tier 1; Armijo preserved exactly), and hoists the invariant $\|g\|^2$ out of the trial loop. Measured effect: mBT fell to 1.93 at $N = 50$ — the dominant contributor to the speedup, because eliminated rejected trials remove both their energy *and* their projection evaluations.

Change B — projection inner-loop cleanup (audit #4). The constant coupling matrix $C = \|v\|^2(I - J/n)$ and its ε-regularized solve are now built once per call rather than per inner iteration; the $\mathcal{O}(VN)$ `A_prev = A.copy()` plus `np.allclose` stall check is replaced by a scalar consecutive-plateau test on `max_error` (gated by `max_error` $< 10\,$tol so it never returns an infeasible iterate); the dead `convergence_history` dict is removed. Result-preserving. Measured effect: the projection's share of wall fell from 88–90% to 80.5%.

Change C — post-step gradient reuse (audit #6). The post-step gradient $g_{\text{post}} = \nabla E_\varepsilon(x)$ is bit-identical to the next iteration's $\nabla E_\varepsilon(x)$ (x is unmodified between them), so it is carried forward, halving gradient evaluations. Bit-identical; $\sim 1.6\%$ wall.

7 What Remains to Scale to $N > 100$

Equation (1) identifies three independent factors in T. The serial optimizations above shrank the constant but left $\alpha \approx 3.1$ intact; reducing the *exponent* requires the algorithmic work below.

1. Further cut re-projection on rejected backtrack steps (partially done). Change A's Tier 1 warm-start already brought mBT from the 4–12 thrashing regime down to ~ 1.9, capturing most of the easy gain. Two stronger options remain, both still future work:

- **Gradient-step rule without line search** (Barzilai–Borwein or spectral step, audit Tier 2): project exactly once per major iteration, removing the residual (mBT+1) multiplier entirely. Non-monotone; needs a Grippo-type safeguard and clamping against the $[10^{-8}, 1 - 10^{-8}]$ box.
- **Lazy projection**: evaluate the unconstrained gradient step first and project only when the Armijo condition is about to be accepted.

With mBT already near 2, the remaining headroom here is modest ($\lesssim 2\times$) — the larger structural wins are below.

2. Replace the iterative projection with a dual-Newton closed-form solve (highest-merit algorithmic fix, audit #3). The coupled sum-to-one + equal-area projection is already closed-form per inner iteration; only the $A \geq 0$ nonnegativity forces the 13–20 inner sweeps. Dropping the area constraint lets each vertex row project onto the probability simplex in closed form ($\mathcal{O}(N \log N)$); the N equal-area couplings are then re-introduced via a dual $y \in \mathbb{R}^N$ solved with a semismooth Newton iteration ($\sim$5–15 steps). This replaces the clamp/Sinkhorn sweeps with ~ 10 Newton steps and gives *exact* feasibility, removing the ε-floor pathology. It is the durable fix for $N \geq 100$ and belongs in its own plan; it was *not* landable before the multi-day production runs.

Remark 7.1 (Inner-solver dual warm-start was ruled out)**.** An earlier draft of this document listed "warm-start the projection inner solver" (carrying dual multipliers between consecutive PGD projections) as a $\sim 2\times$ lever. The serial-optimization audit reclassified this as a **negative finding** (audit #13): the projection's per-call dual state is not a safe warm start across PGD iterates — it risks returning an infeasible iterate — and Change B's measured profile shows mPII is already only ~ 14, leaving little to recover. The structural way to attack mPII is the dual-Newton reformulation above, not warm-starting the existing fixed-point loop.

3. Parallelise across regions. The $\mathcal{O}(N)$ factor in the per-call cost arises from processing each region's density column independently. These columns are embarrassingly parallel: distributing the $V \times N$ density matrix across P cores reduces per-call cost to $\mathcal{O}(VN/P)$, converting the N factor from a wall-clock penalty to a throughput one.

4. Reduce the iteration-count growth exponent. The empirical $K \propto N^2$ trend reflects the increasing difficulty of the energy landscape. Better initialisation (the seeded Voronoi approach already helps at $N = 30$ and $N = 50$), coarser-to-finer scheduling, or a larger initial ε could reduce this exponent. Quantifying the iteration-count scaling precisely requires profiled runs at $N \in \{10, 20, 75\}$ with the seeded init; only the $N = 30$ and $N = 50$ points exist so far.

5. FEM assembly is not the bottleneck. Assembly is $< 1\%$ of wall time at current mesh sizes; it becomes relevant only above $\sim 10^5$–10^6 vertices.

Remark 7.2 (Cluster scaling)**.** Running multiple seeds in parallel on a cluster (e.g. Pelle) does not reduce the per-seed wall time but gives the ensemble result sooner. For $N = 100$ ($\sim$4 days per seed) this compresses the effective turnaround when several seeds are needed. For $N = 1000$ it is not a solution: the per-seed runtime is still measured in years.

Summary table

Factor	Current scaling (post-opt)	Path to improvement
Cost per projection call	$\mathcal{O}(\text{mPII} \cdot VN)$; mPII $\approx$ 14; projection 80.5% of wall at $N = 50$	Dual-Newton closed-form simplex projection (audit #3); parallelise regions ($P\times$)
# projection calls	$K \cdot (\text{mBT} + 1)$; mBT $=$ 1.93 after Change A (was 4–12)	BB/spectral step or lazy projection: remove residual mBT+1 multiplier
Iteration count K	$\sim N^2$ empirically; $K =$ 54k at $N = 50$	Better init, adaptive ε; needs data at $N \in \{20, 75\}$
Overall exponent α	$\approx$ 3.1; unchanged by Changes A/B/C (constant-factor)	Target $\alpha \lesssim 2$ via the dual-Newton projection
Serial opts (A/B/C)	Landed (`bae8cd2`); $\approx$ 3.8$\times$, output-identical	Done — constant-factor only
$N = 50$ (measured)	11.6 h clean host (was projected 100–190 h pre-opt)	Routinely feasible on one machine
$N = 100$ estimate	$\sim$99 h ($\sim$4 days) on one clean machine	Feasible now; cluster for multi-seed
$N = 1000$ estimate	$\sim 1.25 \times 10^5$ h ($\sim$14 yr) on one machine	Requires the algorithmic fix + massive parallelism

References